

Café-au-Lait Spots

Café-au-lait spots may appear darker when overlapping due to increased pigment concentration. When present within Mongolian spots, they are typically surrounded by a lighter pigmented halo. The number of café-au-lait spots does not correlate with disease severity in neurofibromatosis type 1 (NF-1), and their location does not predict where neurofibromas will develop.

These macules represent clusters of heavily pigmented melanocytes of neural crest origin in the epidermis. While the melanocytes may contain increased numbers of "giant" melanosomes or melanin macroglobules, these features are not unique to NF-1 and have no diagnostic value in biopsy specimens.

The diagnostic criterion for NF-1 is the presence of six or more café-au-lait spots greater than 5 mm in diameter in prepubertal individuals or greater than 1.5 cm in adults. Among children eventually diagnosed with NF-1, 53% have six or more such spots by age 3 years, and 97% by age 6 years. Light brown macules resembling café-au-lait spots can also occur in red-headed children, particularly of Irish descent; however, these children usually have fewer than six macules and lack other signs of NF-1. Additionally, infants with multiple lentiginosis syndrome or LEOPARD syndrome (characterized by multiple lentigines, electrocardiographic abnormalities, ocular hypertelorism, pulmonary stenosis, genital abnormalities, growth retardation, and deafness) may present with café-au-lait macules at birth or during infancy, with lentigines appearing later in childhood.

Intertriginous Freckling

Small café-au-lait spots less than 5 mm in diameter are termed freckles and commonly appear in the axillae, inguinal regions, and beneath the breasts. Unlike common sun-induced freckles, these lesions are not related to UV exposure and are considered virtually pathognomonic of NF-1 (Crowe sign). By

age 6 years, 81% of children ultimately diagnosed with NF-1 will exhibit intertriginous freckling.

Discrete Neurofibromas (See also Chap. 127)

Neurofibromas are benign nerve sheath tumors composed of Schwann cells, mast cells, fibroblasts, and perineural cells. They arise from peripheral nerves and present as discrete masses. Cutaneous neurofibromas protrude slightly above the skin surface or lie just beneath it, often with an overlying violaceous discoloration. They are softer than surrounding tissue and may produce a "buttonholing" sensation when palpated (Fig. 142-3).

Subcutaneous neurofibromas, which develop under the skin or deep in visceral tissues, are generally firmer. Those originating from dorsal root ganglia can extend through neural foramina, compressing the spinal cord and forming a "dumbbell" shape. Cervical subcutaneous neurofibromas may feel like a "beaded necklace" and are frequently mistaken for lymphadenopathy. Fewer than 20% of children under 10 years of age develop cutaneous neurofibromas.